GEOSPATIAL MACHINE LEARNING

Building footprints from aerial imagery

SAM 3 and Esri | Lewisville, Texas | May 2026 imagery

A shared Python workflow prepares imagery, runs or imports predictions, cleans building outlines, and evaluates every method under the same rules. The city-scale workflow processed 21.1 gigapixels at approximately 10 cm resolution.

Reviewed test-area accuracy

The primary cohort contains **468 reviewed buildings of at least 20 m²** in one 700 m square. Both inputs are area-filtered before matching. Settings and predictions were frozen; this evaluation did not rerun inference or retune on the reviewed test labels.

Method	Precision	Recall	F1
Esri, city run	0.965	0.951	0.958
Esri, local raw run	0.936	0.904	0.920
SAM 3	0.828	0.923	0.873

All sizes: a separate evaluation uses 600 reviewed buildings, including 132 outlines under 20 m². Forty boundary-crossing outlines are excluded. Matching is one-to-one at IoU >= 0.5 in EPSG:26914.

Method	All-size F1	Missed reference
Esri, city run	0.839	155
Esri, local raw run	0.808	172
SAM 3	0.772	164

In the all-size matching, SAM 3 and local Esri each match 4 of the 132 reference outlines under 20 m²; city Esri matches none. These area classes do not identify building types.

Scope: these results describe one reviewed test square. They do not establish city-wide accuracy. Changes from historical scores reflect the changed reference and scoring policy, not an improvement in the models.

Historical agreement stays separate

Earlier city-wide scores use 38,882 historical outlines from a 38,924-record source inventory. Dates are primarily 2015 or earlier; 33 source records are undated and remain in the evaluated set. These scores measure historical agreement, not current accuracy.

Method	City-wide F1, at least 20 m²
Esri, city run	0.834
Esri, local raw run	0.766
SAM 3	0.744

An outline absent from the inventory is a candidate for review. Absence does not establish a construction date or prove that a prediction is a building. The city and local Esri outputs have different processing histories.

Review and engineering

Human review: Esad completed both review areas and confirmed the labeling rules and no unresolved cases. The test review used only aerials and historical outlines, with no prior or concurrent prediction exposure. These are reviewer attestations. The reference retains inherited geometry and splitting conventions; no second independent reviewer is recorded.

Workflow: hash-linked tile manifests, resumable inference, chunk ownership for ArcGIS execution, shared cleanup, explicit boundary handling, and one-to-one scoring. Automated tests cover Windows and Ubuntu. Tile overlap reduces edge losses; buildings longer than a tile can still be split.

Publication: committed aggregates, charts and provenance support report regeneration. Raw imagery, reviewed labels, prediction geometry and full receipts remain local. The full report retains its previously published screenshots. Public files alone cannot reproduce the geometric evaluation.

Completed baseline; fine-tuning deferred

Reference review (#2) and frozen evaluation (#3) are complete following merged PR #29. Remaining issue #4 requires separate training, validation and fresh test areas, with every eligible building labeled and no prediction exposure during fresh test review. Preserve the completed test reference and scores. Labeling packages for #4 have not been prepared. Training feasibility on the 6 GB GPU remains unverified.

Public evidence and guides

Repository and setup

Reviewed scores, protocol and provenance

Full comparison report, 11 pages

Usage and reproduction guide